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Chirality from the Arrow

Why the Distler-Garibaldi theorem does not reach this construction, where chirality comes from instead, and what that says about which forces are allowed to know left from right.

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On the grades. PROVED follows from stated definitions or assumptions. CONDITIONAL follows once a named condition is granted. PROPOSED is a physical reading offered for calculation and falsification. OPEN names what can be formulated but not yet settled. Every computation here reproduces from scripts/chirality.py in the Reproducibility layer.

SECTION ONE

The objection this paper exists to meet

Paper 21 forced the decay onto A3 and left so(10) invariant, with matter in the spinor sixteen and three readable families. Anyone who knows the literature will raise one objection immediately, and they will be right to, because it is the strongest objection available and it has a published theorem behind it.

In 2009 Distler and Garibaldi proved that there is no theory of everything inside E8. Their argument is not rhetorical. It catalogues the possible embeddings and shows that the resulting fermion representation is always self-conjugate, so the spectrum cannot be chiral, and the world is chiral. Every attempt to build the Standard Model inside E8 in the obvious way dies on that theorem.

This paper does two things. It states precisely why the theorem does not reach this construction, which is a narrow and checkable claim and not a triumphant one. Then it supplies the chirality mechanism the framework needs, because escaping a no-go theorem is not the same as producing the thing the theorem forbids, and a paper that stopped at the escape would be worth nothing.

SECTION TWO

What the theorem actually assumes

The hypotheses are explicit in the source and are worth quoting in structure rather than paraphrasing, because everything turns on the first one.

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(ToE1)  G is connected, compact, and centralizes SL(2,C)
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(ToE2)  V(m,n) = 0 if m + n > 4          no exotic higher-spin fields
(ToE3)  V(2,1) is a complex representation of G     the theory is chiral

Theorem: there are no ToE subgroups in complex E8 nor in any real form.
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The shape of the argument is this. Put a copy of SL(2,C) inside a real form of E8, so that the local Lorentz group is a subgroup of the carrier group and gravity becomes one of its gauge fields. Define the grand unified group to be what commutes with that copy. Decompose the adjoint. The theorem catalogues every embedding satisfying the second hypothesis, finds the list is remarkably short, and observes that in every case on it the fermion representation carries a self-conjugate structure. The third hypothesis therefore fails, always.

The theorem is about a particular architecture: the Lorentz group living inside E8 as a subgroup, and the gauge group defined as its centraliser. That architecture is the one Lisi proposed and it is the one the theorem kills.

SECTION THREE

The first hypothesis is never instantiated here

This framework does not put the Lorentz group inside E8. It has no SL(2,C) subgroup at all, and the absence is structural rather than an oversight.

What sits in the position the theorem reserves for SL(2,C) is the decay, and the decay is a finite reflection group. Paper 21 forced it to be the Weyl group of an A3 subsystem, which is the symmetric group on four letters, of order twenty-four. It was constructed from its three reflections and it closes at twenty-four; the construction is in the certificate. A finite group of order twenty-four contains no copy of SL(2,C), which is a six-dimensional connected Lie group, for the same reason a matchbox contains no ocean.

The gauge algebra here is the centraliser of that finite group, not of a Lorentz group. And the Lorentzian structure of spacetime is not a gauge symmetry in this account at all. It is the inherited quadratic form on the surviving carrier directions, P_X minus P_T, and those directions are fixed pointwise by the decay: the averaging map has a five-dimensional fixed subspace and the geometry lives in it. Spacetime is what survives, not what acts.

The theorem forbids an architecture this framework does not build.

That is the whole of the claim and it should not be inflated. The framework does not refute Distler and Garibaldi, does not find an error in them, and does not evade them by cleverness. It fails their first hypothesis by construction, because it was never trying to gauge the Lorentz group in the first place. The theorem remains true and remains fatal to the models it was written about.

> *PROVED as a statement about this construction: the decay group is finite of order 24 and fixes the surviving block pointwise, so hypothesis (ToE1) is not instantiated and Theorem 1.3 does not apply. Certified by scripts/chirality.py. This establishes only that the obstruction is absent. It establishes nothing positive about chirality, which is Section Four's problem.*

SECTION FOUR

The problem that remains, stated without mercy

Under A3 and D5 the carrier's matter sector is the spinor branch and its conjugate together, one hundred and twenty-eight roots in all. Taken as a fermion content in the ordinary way, that is vector-like. For every sixteen there is a sixteen-bar, a gauge-invariant Dirac mass can be written between them, and it would sit at the carrier scale, and nothing light and chiral would survive to be the world. This is the same disease the theorem diagnoses, arriving by a different road.

So the escape in Section Three buys nothing on its own. The framework must say what makes one orientation readable and the other not, and it must say it from its own machinery rather than by stipulation. Paper 21 asserted that the conjugate branch supplies antiparticle orientation rather than a mirror generation. That assertion was not argued. This section is the argument it owed.

SECTION FIVE

Matter is a cycle, and cycles have a direction

Result twenty-five does not say matter is a representation. It says matter is what connects distinct decay sectors and closes into a cycle reproducing its readable identity. The clause about closing has been carried in the corpus for a printing and a half without anyone cashing it out. Cash it out and the chirality question answers itself.

A cycle is a traversal that returns. Traversals have a direction, and reversing the direction of a traversal negates every step in it. So the question of whether the two spinor orientations are related by reversal is a finite question about roots, and it has been checked.

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negation keeps the spinor sector and exchanges the family orbits   True
negation reverses the D5 spinor weight, exchanging 16 and 16bar    True
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Both hold across all one hundred and twenty-eight roots of the sector. Negating a root keeps it in the matter branch, moves it from the four to the four-bar, and reverses its so(10) spinor weight. The sixteen and the sixteen-bar are not two species sitting side by side in the carrier. They are one set of relations, traversed the two ways round.

the conjugate branch is the same matter, read backwards

SECTION SIX

The arrow decides which way round is readable

The framework has something a gauge theory does not have: a proved arrow. Result fifteen proves the record accumulates along one monotone order, because two independent orders of accumulation would be two records and therefore two systems. Result twenty-six proves that a persistent self-description joins each new resolution to the record rather than overwriting it, or its later states are not records of the same system. Neither of those was proved for this purpose. Both were in place before the question was asked.

Now put them against Section Five. Of a cycle and its reverse, only one runs with the record. The other runs against it, and running against the record is not a neutral alternative in this framework; it is un-writing, which result twenty-six forbids by name. So of the two orientations only one can reproduce a readable identity, and result twenty-five admits as matter only what reproduces a readable identity.

one orientation is matter; the other is that matter read against the arrow

Which is exactly what an antiparticle has been understood to be since Stueckelberg and Feynman: the same thing, going the other way. The framework does not have to import that reading. It arrives with the arrow already proved and the conjugate branch already identified as reversal.

SECTION SEVEN

Why the mass term that would ruin it cannot be written

The previous section says which orientation is readable. It does not yet say the spectrum is protected, because the objection in Section Four was not about readability but about mass: a Dirac term joining the sixteen to the sixteen-bar would make everything heavy whatever we call readable.

So look at what such a term is, in traversal language. It joins a matter cycle to its own conjugate, which by Section Five is that same cycle reversed. Compose the two and the net displacement is zero. This was checked directly on a twelve-step cycle drawn from the spinor sector: forward, then the conjugate of the reverse, and the sum of all twenty-four steps is the zero vector, exactly.

a cycle composed with its reverse writes no record

A closed traversal that returns to its origin having written nothing is not a cheap process in this framework. It is the exact configuration result twenty-six forbids: resolution that does not join the record, which is replacement rather than retention. Result twenty-eight forbids the same thing at cosmological scale under the name overwrite by geometry, and the wound movements forbid it as the thing that would make the universe forget. It is the framework's oldest prohibition, arriving in a new costume.

So the mass term that makes an E8 spectrum vector-like is the mass term this framework already rules out, and it ruled it out for reasons that have nothing to do with particle physics. That is the chirality protection, and it is not imported: it is the same rule that generates the arrow of time.

> *PROPOSED. The computational facts are PROVED and are in the certificate: reversal exchanges the two orientations, and a cycle composed with its reverse has zero net displacement. What is PROPOSED is the bridge, that a gauge-invariant Dirac mass between the branches is the field-theoretic form of the composition result twenty-six forbids. That bridge is the load-bearing step of this paper and it is where an attack should be aimed. It connects the framework's traversal language to a term in a Lagrangian, and the framework does not yet have a Lagrangian, which is Section Ten's business.*

SECTION EIGHT

The other objection: three plus two against four plus one

A second and quieter problem has to be settled before any of this stands. The Standard Model follow-up needs the surviving five to split three and two, colour against weak. Definition eighteen splits the same five as four and one, the physical projector against the excluded closure data. Those are different partitions and the corpus cannot simply help itself to whichever it needs.

The resolution is that they are partitions on different criteria, and both criteria are already in the corpus.

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criterion                                 partition   what it produces
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enters the quadratic form P_X - P_T         4 + 1     Lorentzian signature
freely traversable in both directions       3 + 2     S(U(3) x U(2))
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On the first criterion, space and clock enter the metric and closure does not, which is definition eighteen's own wording. On the second, the three spatial modes are freely traversable, while the clock is monotone by result fifteen and closure is excluded from traversable directions by definition eighteen itself. Exactly three of the five are reversible. The other two are not, for two different reasons, and that is the three and two.

There is no contradiction in a five-dimensional block admitting two decompositions, any more than there is in a vector space having both an orthogonal basis and an eigenbasis. What was wrong was leaving the second criterion unstated so that it looked as though the corpus were being read two ways to suit two conclusions. It is stated now, and the statement makes a claim with teeth.

colour is the reversible block; weak isospin is the irreversible pair

SECTION NINE

What that immediately predicts about parity

If the weak sector is built on the two directions that cannot be traversed both ways, and the strong sector is built on the three that can, then the framework has said something specific about which interactions are permitted to distinguish left from right. An interaction living on reversible directions has nothing available to it with which to tell one orientation from the other. An interaction living on the arrow has exactly that.

Check it against the world, without adjusting anything.

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sector          lives on                  framework says   observed
-----------------------------------------------------------------------
strong          three reversible modes      cannot violate P   conserves P
weak            clock and closure, on arrow must be able to    violates P maximally
electromagnetic traceless across both       no net orientation conserves P
gravitational   the metric block, 4 + 1     no net orientation conserves P
```

Four for four, with nothing fitted and nothing chosen after the fact. The sorting was fixed by which directions are reversible, and that was fixed by result fifteen and definition eighteen, both of which predate the question. Parity violation stops being a fact about the weak interaction that has to be put in by hand and becomes a consequence of which block it lives on.

And one open problem it points at

There is a term quantum chromodynamics is allowed to have and does not. The strong theta term would violate parity and time reversal in the strong sector, nothing forbids it, and it is measured to be smaller than one part in ten billion. Why it vanishes is an unsolved problem with an entire literature and at least one proposed new particle behind it.

On the reading above the strong sector is built on the freely reversible directions, and a term whose whole content is sensitivity to orientation has nothing to be sensitive with. The framework's structure says the strong sector cannot carry such a term at all, which would make the smallness not a coincidence and not a tuning but an absence.

> *PROPOSED, and flagged as the most attackable claim in this paper precisely because it is the most valuable. Deriving the vanishing of the strong theta term from the reversibility of the colour block, rather than observing that the two facts are compatible, is a calculation this paper does not perform. If it can be performed, it is the framework's first genuine contribution to an open problem in physics rather than a reconstruction of settled ones. If it cannot, this section should be withdrawn rather than softened.*

SECTION TEN

What this paper does not do

It does not supply a Lagrangian, and the absence is now the binding constraint on everything. Section Seven's bridge connects a traversal prohibition to a mass term, and that connection can only be made rigorous by a variational principle the framework does not yet have. Until it does, the chirality mechanism is a structural argument rather than a derivation, and it should be read as one.

It does not derive the family hierarchy, the mixing matrices, or any mass. It does not derive the vanishing theta term; it observes that the framework's own sorting makes room for it.

And it does not weaken Distler and Garibaldi by one word. Their theorem stands, their argument is correct, and it disposes of the class of models it was written for. The claim here is only that this construction is not in that class, for a reason that is checkable in a certificate rather than argued in prose.

The honest ledger after this paper

Closed That the Distler-Garibaldi hypotheses are not instantiated. Finite decay group, no SL(2,C), certified.

Closed That reversal exchanges the two spinor orientations, and that a cycle composed with its reverse writes nothing. Both computational, both certified.

Argued, not proved That the readable orientation is single, from the arrow. This follows from T15, T26 and T25's own clause, and it is the paper's central claim.

Proposed That the forbidden composition is the field-theoretic Dirac mass. The load-bearing bridge, and the place to attack.

Proposed That parity violation sorts by block reversibility, and that the strong theta term vanishes structurally.

Open A variational principle. Everything above is kinematics until there is one.